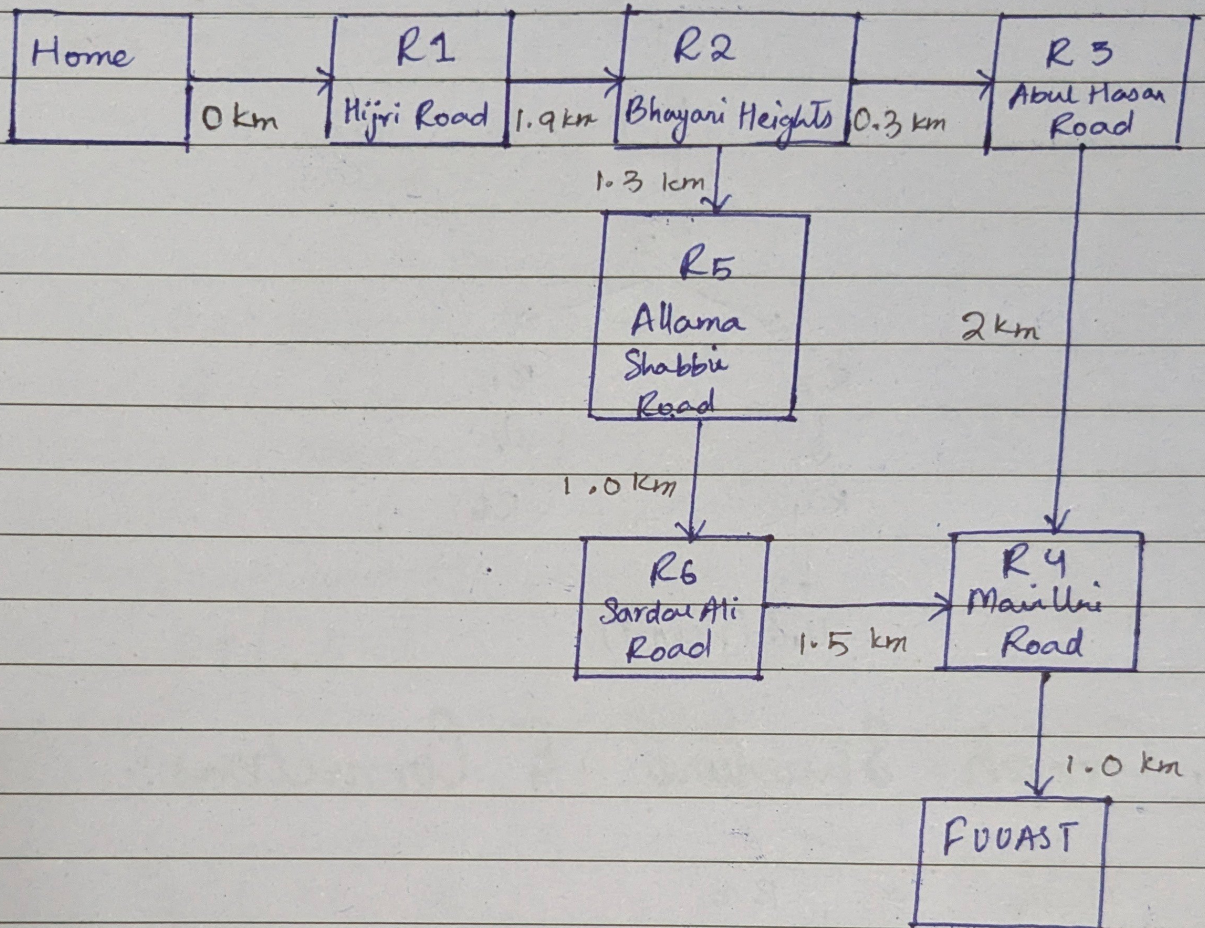


BFS: From Home to Uni

Date _____

• Map :-



Path 1 :-

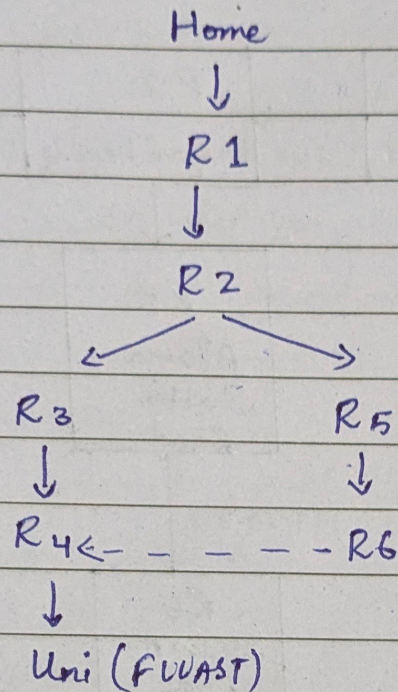
$R_1 \rightarrow R_2 \rightarrow R_3 \rightarrow R_4 \rightarrow \text{FUUAST}$

$$0 + 1.9 + 0.3 + 2 + 1.0 = 5.2 \text{ km}$$

Path 2 :-

$$0 + 1.9 + 1.3 + 1.0 + 1.5 + 1.0 = 6.7 \text{ km}$$

• Free Structure :-



• Graph Structure & Connections :-

Home → R1

R1 → Home, R2

R2 → R1, R3, R5

R3 → R2, R4

R4 → R3, R6, Uni

R5 → R2, R6

R6 → R5, R4

Uni → R4

(FLUAST)

. BFS Trace :-

Step 0 :- Initialization

Queue : [Home]

Visited : [Home]

Home

Step 1 :- Dequeue, & expand

Q : [R1]

v : [Home, R1]

Step 2: Dequeue R1 and Expand

Q : [R2]

v : [Home, R1, R2]

Step 3: Dequeue R2 and Expand

Q : [R3, R5]

v : [Home, R1, R2, R3, R5]

Step 4: Dequeue R3 and Expand

Q : [R5, R4]

v : [Home, R1, R2, R3, R5, R4]

Step 5: Dequeue R5 and Expand

Q : [R4, R6]

v : [Home, R1, R2, R3, R5, R⁴~~5~~, R6]

Step 6: Dequeue R4 and Expand

Q: [R6, Uni]

V: [Home, R1, R2, R3, R5, R4, R6, Uni]

Step 7: Dequeue R6 and Expand

Q: [Uni]

V: [Home, R1, R2, R3, R5, R4, R6, Uni]

Step 8: Dequeue Uni

Goal Found!

• BFS ORDER OF EXPANSION :-

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R5 $\rightarrow$ R4 $\rightarrow$ R6 $\rightarrow$ Uni

• SHORTEST PATH FOUND (HOME $\rightarrow$ UNI) :-

Home $\rightarrow$ R1 $\rightarrow$ R2 $\rightarrow$ R3 $\rightarrow$ R4 $\rightarrow$ Uni